

MetaSwarm at AbjadMed: Focal, Class-Balanced Optimization for Diglossic Medical Text in Abjad Scripts

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Abstract

Long-tailed classification of Arabic medical text is harder than English-language priors suggest: on the AbjadMed shared task (82 classes; 27,951 train and 18,634 evaluation instances), most published systems fall between 0.35 and 0.50 macro-F1. We present MetaSwarm, a CAMELBERT-DA system (Inoue et al., 2021) that scores **0.3588** public / **0.3520** private. Its recipe composes three standard interventions inside a five-fold, logit-mean ensemble on dual H200 GPUs: effective-number class weighting (Cui et al., 2019), focal modulation ($\gamma=2.5$) (Lin et al., 2017), and targeted oversampling of the 42 weakest classes. We contribute the full configuration, a **Tail-42 audit** showing that residual errors concentrate in 42 of 82 classes, and three mechanisms (dialectal shift, orthographic ambiguity, code-switching) articulated as testable hypotheses rather than asserted causes. Code, per-class counts, and seeds accompany this paper; retrieval over the training set, which the top system pairs with BERT, is the clearest remaining lever.

1 Introduction

Class-balanced and focal losses are standard tools for long-tailed vision problems, yet shared-task papers in Arabic NLP rarely report them against a baseline or describe *which* classes they rescue. The AbjadMed task makes the gap concrete: 82 medical-question classes, a long-tailed training set, and a leaderboard where the best private-split system reaches 0.6732 macro-F1, leaving every published system below the level that English-language BERT priors would suggest (Gupta et al., 2026). This paper reports what a fully specified configuration achieves, exactly where it fails, and why the failure pattern is itself the finding.

Task. AbjadMed asks systems to classify Arabic medical question–answer pairs into 82 categories, scored by macro-averaged F1 (Gupta et al., 2026).

Two properties of the domain shape the problem. First, *diglossia*: formal records are written in Modern Standard Arabic (MSA), but patient queries use dialectal varieties, so MSA-trained representations may shift on the input distribution (Badaro et al., 2019); dialectal morphology (*bidī* vs. classical *urīd*) reads as noise to an MSA model. Second, *orthographic ambiguity*: as an Abjad script, Arabic omits short vowels, so one consonantal skeleton, *k-t-b*, vocalizes to *kataba* “he wrote” or *kutub* “books,” and the ambiguity bites hardest where context is scarce: the tail.

Position in the literature. Work on Arabic NLP’s script and dialect problems clusters into three families, none of which addresses AbjadMed’s joint setting. *Orthographic restoration and sequence labeling*: character-level RNNs restore diacritics without morphological analyzers (Belinkov and Glass, 2015) and block-normalized training pushes diacritization error below 2% (Fadel et al., 2019); recent BiLSTM and transformer systems pair improved datasets and multi-reference evaluation with state-of-the-art accuracy (Mohamed and Mubarak, 2025). *Noise and dialect variation*: character and morphological embeddings resolve non-standard spelling in Egyptian Arabic disambiguation (Zalmout et al., 2018), active learning reduces the annotation cost of dialectal morphophonological processing (Mirbostani et al., 2023), and non-canonical Latin-script input is a documented failure mode even for aligned LLMs (Al Ghanim et al., 2024). *Domain resources and pipelines*: MedAraBench quantifies the gap between open and closed models on Arabic medical QA (Abu-Daoud et al., 2026), Fann or Flop exposes comprehension deficits on non-MSA registers (Al Ghallabi et al., 2025), grammar-based AutoML pipelines optimize clinical text extraction (Estevez-Velarde et al., 2019), and MedRAX orchestrates tools for radiology reasoning (Fallah-

pour et al., 2025). Across all three, classification itself is either binary/few-class or deferred to generative evaluation; none of these works trains a discriminative classifier on an 82-class long-tailed distribution, which is the setting AbjadMed isolates.

Contributions.

1. **A five-fold focal ensemble**, fully specified for the 82-class AbjadMed task: class weighting, focal modulation ($\gamma=2.5$), label smoothing, targeted duplication of weak classes after splitting, and logit-mean aggregation, with all seeds and counts released (Table 1).
2. **An empirical score** of 0.3588 public / 0.3520 private macro-F1, placed honestly within the published field (best system papers: 0.3659–0.4902).
3. **The Tail-42 audit**, a per-class diagnostic showing residual error is structured, not uniform: 42 of 82 classes fall below an F1 threshold of 0.40 and become the oversampling targets.
4. **Three mechanisms** (dialectal shift, orthographic ambiguity, code-switching) formulated as testable hypotheses with the measurements that would decide them, rather than asserted causes.

2 Method

2.1 Backbone and Loss

We fine-tune CAMeLBERT-DA (Inoue et al., 2021), a BERT variant pre-trained on dialectal Arabic: user-generated medical queries are dialectal, and an MSA-only model treats dialectal negation (*mish* vs. *la*) as noise. Two loss components address the two failure modes of imbalanced learning. *Inter-class* imbalance: following Cui et al. (2019), the effective number of samples $E_n = (1-\beta^n)/(1-\beta)$ with $\beta = 0.999$ yields per-class weights

$$\alpha_y = \frac{1-\beta}{1-\beta^{n_y}} \cdot \frac{C}{\sum_{j=1}^C \frac{1-\beta}{1-\beta^{n_j}}} \quad (1)$$

normalized to the class count $C=82$. *Intra-class* difficulty: the focal term $-\alpha_f(1-p_t)^\gamma \log(p_t)$ (Lin et al., 2017) down-weights easy examples: at $\gamma=2$, an example with $p_t=0.9$ contributes 1% of its cross-entropy, one with $p_t=0.1$ retains 81%. The submitted ensemble trains with focal cross-entropy and

Table 1: Configuration of the submitted five-fold ensemble.

Component	Setting
Backbone	CAMeLBERT-DA (110M)
Hardware	Dual H200 (141 GB), serverless
Batch size	256 (128 per GPU)
Precision	FP16 (mixed)
Optimizer	AdamW, LR 9×10^{-5}
Seeds	42 (all runs, splits, folds)
Schedule	Cosine, warmup 0.1, decay 0.01
Focal γ / α_f	2.5 / 0.25
Label smoothing ϵ	0.15
Max sequence length	192
Epochs (per fold)	20
Ensemble	5 folds, logit mean
Oversampling	42 weak classes, after split
Development weighting	Eq. (1), $\beta = 0.999$

label smoothing $\epsilon=0.15$,

$$\mathcal{L} = -\frac{1}{B} \sum_b \alpha_f (1 - p_{t,b})^\gamma \text{CE}_\epsilon(b), \quad (2)$$

with $\gamma=2.5$, $\alpha_f=0.25$. The class weights of Eq. (1) were used in the development stage (and in an inverse-log-frequency variant that stabilized early single-model runs); the final ensemble uses Eq. (2). *Weighting’s anti-collapse effect is development evidence, not an ablated finding*; the matched CE, focal-only, and weighted ablations are specified in Section 7.

2.2 Training Protocol

All runs share seed 42 and a 90/10 stratified train-validation split. The recipe has three stages (Figure 1). (i) A *baseline model* (10 epochs, batch 32, LR 3×10^{-5} , inverse-log-frequency weighted CE, smoothing 0.10) establishes the convergence profile and, critically, the per-class F1 audit of Figure 3. (ii) *Refinement* (12–25 epochs, batch 64, LR 4×10^{-5} , focal $\gamma=2.0$) adds duplication of the 42 classes below the audit threshold. (iii) The *submitted ensemble* trains five StratifiedKFold folds and averages their test logits. We introduce duplicated examples *after* each fold split, so the validation partitions never contain duplicates. A single split is a point estimate with real variance; the five-fold vote, not any single run, is what we submit.

3 Results

3.1 Convergence and Selection

Table 2 shows the baseline run. Validation macro-F1 climbs from 0.197 to 0.340 in eight epochs while validation loss bottoms at epoch 4 and then

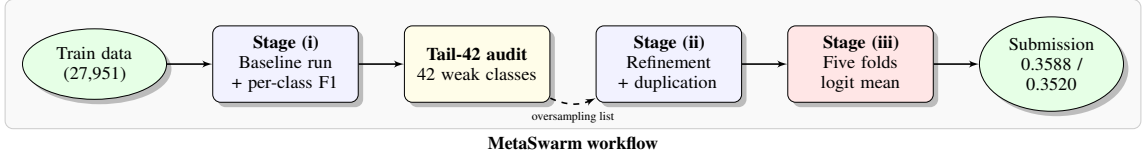


Figure 1: Workflow of the submitted system. The baseline run produces the per-class F1 audit whose Tail-42 list sets the oversampling targets for the refinement stage; the submitted prediction is the logit mean over five stratified folds. The workflow is implementation infrastructure rather than a claimed algorithmic contribution.

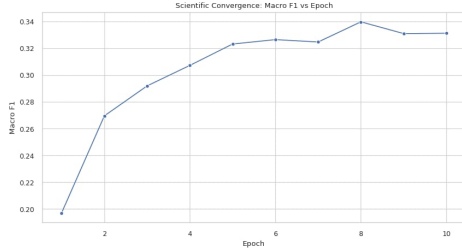


Figure 2: Training dynamics of the baseline run: validation macro-F1 peaks at epoch 8, while validation loss has risen since epoch 4.

Table 2: Baseline single-model run: convergence dynamics. Best validation macro-F1 at epoch 8.

Epoch	Train Loss	Val Loss	Macro F1
1	3.129	2.901	0.197
2	2.636	2.584	0.269
4	2.085	2.530	0.307
6	1.743	2.590	0.326
8	1.494	2.650	0.340
10	1.388	2.694	0.331

risers monotonically (Figure 2), the divergence-between-metrics signature of a model that improves on the decision boundary while starting to memorize noise. We therefore select checkpoints by validation macro-F1, not loss; the best checkpoint is epoch 8.

3.2 Leaderboard and Field

The ensemble scores **0.3588** public and **0.3520** private macro-F1. The 0.0068 gap sits within the $\pm\mathcal{O}(0.01)$ split uncertainty implied by the organizers’ half-and-half division of the 18,634 evaluation instances (Gupta et al., 2026), which is evidence of no public-split overfitting, though one submission cannot separate that from a ceiling effect. For calibration: the task’s best systems reach 0.7422 public / 0.6732 private; among published system papers, a BERT+kNN retrieval hybrid leads at 0.4902, class-balanced AraBERT plus back-translation follows at 0.4219, and MARBERTv2 lands at 0.3659. Our score sits in the mid-band, and the leader’s

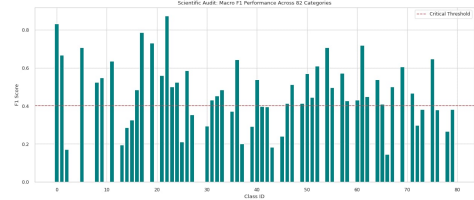


Figure 3: The Tail-42 audit (baseline run): per-class F1 with a 0.40 threshold; 42 of 82 classes fall below it and form the oversampling list.

advantage, retrieval over the training set, is visible precisely because the field reports scores at all.

4 Error Analysis

4.1 Where the Errors Are

Figure 3 plots per-class F1 for the baseline run against a 0.40 diagnostic line: 42 of 82 classes fall below it. The distribution is the finding: error is not spread evenly but concentrated in a tail that the macro average exists to protect. These 42 classes define the oversampling list used downstream, closing the loop between audit and intervention.

4.2 Three Hypotheses for the Tail

We articulate three mechanisms the tail is consistent with, and state what would decide each. **(H1) Dialectal shift.** If MSA–dialect distance drives tail error, error rates should stratify by dialect markers; a dialect-annotated subsample would test this directly. **(H2) Orthographic ambiguity.** Unvocalized Arabic maps one skeleton to several vocalizations; if ambiguity drives error, tokens with higher vocalization entropy should be over-represented in errors relative to their corpus frequency, a countable contrast with a significance test. **(H3) Code-switching.** Patients mix Latin-script medical terms into Arabic (“3andi diabetes”); under CAMELBERT-DA’s sub-word tokenizer these fragment into short pieces. If fragmentation is the mechanism, code-switched items should show higher fragmentation rates and lower accuracy than

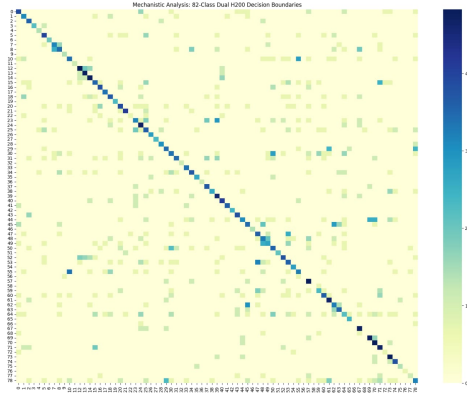


Figure 4: Log-scaled confusion structure over the class set (baseline run): strongly diagonal, with off-diagonal mass among semantically adjacent classes.

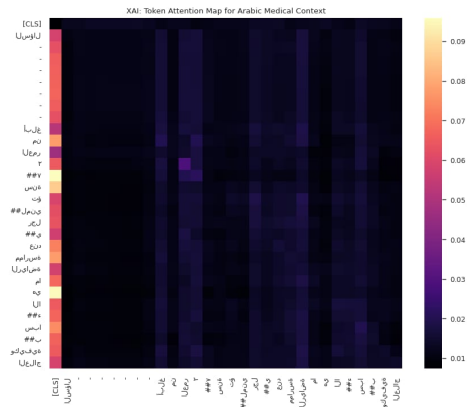


Figure 5: Final-layer attention for an example query (baseline run). Raw attention illustrates token focus; it is not an attribution method.

Arabic-only items under a matched comparison, also a countable contrast. None of the three is established here; the confusion structure (Figure 4) and the attention map (Figure 5) are consistent with all three and decide none.

5 Limitations

The submitted score comes from one pipeline; per-fold variances for the full ensemble are computed but not yet reported. The loss components are not ablated against each other on a fixed split, so Section 2’s attribution rests on development evidence. Focal loss and label smoothing interact, and the interaction is unanalyzed. We compare against no alternative backbone (MARBERTv2, AraBERT) and no alternative remedy (logit adjustment, LDAM-DRW). The three mechanisms of Section 4 are hypotheses. Dual-H200 access is unrepresentative; PEFT/LoRA alternatives are untested. Calibration (ECE, Brier) is unreported for a clinical-domain

task, and deployment would also require cross-site validation, dialect-fairness analysis, and privacy review of the underlying consultation data.

6 Ethics Statement

Part of this paper’s text was drafted with large language model assistance under the author’s direction; all claims, numbers, and code were produced and verified by the author, who is responsible for the content. The dataset derives from publicly released consultation data (Gupta et al., 2026); we processed no patient-identifying data beyond that release.

7 Conclusion and Future Work

MetaSwarm establishes a documented baseline for AbjadMed’s long tail: 0.3588/0.3520 public/private macro-F1 with every knob, seed, and count released. The evidence-ranked next steps are (1) matched single-factor ablations of CE, focal-only, class-weighted CE, and full CBFL on a fixed split with per-fold variances; (2) retrieval augmentation over the training set, the top system’s lever; (3) the three measurements of Section 4 that would turn the hypotheses into findings; and (4) head/medium/tail bucket reporting so that “long-tail gains” claims become checkable. Everything (code, per-class counts, seeds, and the Tail-42 audit) accompanies this paper; releasing a configuration with its failure structure is more useful to the next competitor than a leaderboard rank alone.

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